

Reproducing Cardiac Restitution Properties Using the Fenton–Karma Membrane Model

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Abstract—Modern models of cardiac membranes are often highly complex and computationally expensive, particularly when used in long simulations of spatially extended models of cardiac tissue. Therefore, there is a need for simpler membrane models that preserve the features of the complex models deemed important. This communication describes an empirical procedure that was used to choose the parameters of the three-variable Fenton–Karma (FK3V) model to reproduce the restitution properties of the Courtemanche–Ramirez–Nattel model of atrial tissue. The resulting parameter values for the FK3V model and the sensitivity table for all its parameters are provided. Thus, this study gives insight into the behavior of the FK3V model and the effect of its parameters on restitution properties.

Keywords—Cardiac membrane models, Numerical simulations, Atrial tissue.

INTRODUCTION

Modern models of cardiac membranes can have more than 20 state variables and an even larger number of governing equations. Such models are computationally expensive, particularly when used in problems with two or three spatial dimensions. If long periods of time must be simulated, such models can be computationally prohibitive. Therefore, there is tremendous incentive to use less complex membrane models.

Recently, Fenton and Karma developed a generic model of the cardiac membrane.⁵ The model has only three membrane currents: the fast inward current (J_{fast}), the slow inward current (J_{slow}), and the ungated outward current (J_{ung}). The model has only three state variables: the membrane potential (V_m), the fast gate (f), and the slow gate (s). The equations of the Fenton–Karma (FK3V) model are presented in Table 1. For convenience, Table 1 uses the notation adopted by our group, which is different than that used in the original publication.⁵ The parameters of this model can

be chosen so that the model reproduces the action potential duration (APD) restitution and the conduction velocity (CV) restitution obtained from other membrane models or from experiments.⁵ Hence, this simple model can potentially reproduce the features of cardiac tissue considered to play a critical role in cardiac dynamics and arrhythmias.

The primary purpose of this communication is to present a procedure used by our group to choose FK3V parameters based on the APD and CV restitution curves. Our procedure derives from the method originally used by Fenton.⁴ However, our study is more comprehensive as it examines the effect of each FK3V parameter on all restitution features and thus yields greater insight into the behavior of the FK3V model. The original publication of the FK3V model contained parameters for the Beeler–Reuter, modified Beeler–Reuter, and modified Luo–Rudy membrane models.⁵ Therefore, another purpose of our study was to identify parameters for a modern atrial model. Specifically, the procedure was used to choose parameters that match the restitution properties of the Courtemanche–Ramirez–Nattel (CRN) model of atrial tissue.¹

METHODS

Model

The APD and CV restitution curves were generated using a model of a one-dimensional cardiac fiber. The governing equation was

$$\frac{a}{2R_i} \frac{\partial^2 V_m}{\partial x^2} = C_m \frac{\partial V_m}{\partial t} + I_{\text{ion}} - I_{\text{stim}} \quad (1)$$

where V_m is the membrane potential in mV, R_i is the intracellular resistivity ($125 \Omega\text{cm}$), a is the fiber radius ($8 \mu\text{m}$),¹ x is the position along the fiber in cm, C_m is the membrane surface capacitance ($1.99 \mu\text{F}/\text{cm}^2$),¹ t is the time in ms, I_{ion} is the ionic current in $\mu\text{A}/\text{cm}^2$, and I_{stim} is the stimulus current ($134.4 \mu\text{A}/\text{cm}^2$ for 4.5 ms for CRN, and $73.8 \mu\text{A}/\text{cm}^2$ for 4.5 ms for FK3V, 1.25 times diastolic threshold in both cases). The ends of the fiber were considered sealed. The

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TABLE 1. Fenton–Karma membrane model.

Potential equation	$dv/dt = -(J_{\text{fast}} + J_{\text{slow}} + J_{\text{ung}} - J_{\text{stim}})$
Fast inward current	$J_{\text{fast}} = -(f/\tau_{\text{fast}})F(v)$; $v < V_f$: $F(v) = 0$; $v > V_f$: $F(v) = (v - V_f)(1 - v)$
Fast gate equation	$df/dt = (f_{\infty} - f)/\tau_f$ $v < V_f$: $f_{\infty} = 1$, $\tau_f = \tau_{\text{fopen}}$; $v < V_{\text{fopen}}$: $\tau_{\text{fopen}} = \tau_{\text{fopen1}}$; $v > V_{\text{fopen}}$: $\tau_{\text{fopen}} = \tau_{\text{fopen2}}$ $v > V_f$: $f_{\infty} = 0$, $\tau_f = \tau_{\text{fclose}}$
Slow inward current	$J_{\text{slow}} = -(s/\tau_{\text{slow}})S(v)$; $S(v) = 0.5(1 + \tanh(k(v - V_{\text{slow}})))$
Slow gate equation	$ds/dt = (s_{\infty} - s)/\tau_s$; $v < V_s$: $s_{\infty} = 1$, $\tau_s = \tau_{\text{sopen}}$; $v > V_s$: $s_{\infty} = 0$, $\tau_s = \tau_{\text{sclose}}$
Ungated outward current	$J_{\text{ung}} = (1/\tau_{\text{ung}})U(v)$ $v < V_u$: $U(v) = v$, $\tau_{\text{ung}} = \tau_{\text{ung1}}$ $v > V_u$: $U(v) = 1$, $\tau_{\text{ung}} = \tau_{\text{ung2}}$
Parameter values chosen to reproduce APD and CV restitution of CRN atrial model	$V_f = V_s = V_u = 0.160$, $V_{\text{fopen}} = 0.040$, $V_{\text{slow}} = 0.85$, $k = 10.0$ $\tau_{\text{fast}} = 0.249$ ms, $\tau_{\text{fopen1}} = 40.0$ ms, $\tau_{\text{fopen2}} = 82.5$ ms, $\tau_{\text{fclose}} = 5.75$ ms $\tau_{\text{slow}} = 226.9$ ms, $\tau_{\text{sopen}} = 100.0$ ms, $\tau_{\text{sclose}} = 300.0$ ms, $\tau_{\text{ung1}} = 64.7$ ms, $\tau_{\text{ung2}} = 222.9$ ms

FK3V model is formulated in terms of dimensionless voltage v while the CRN model uses membrane potential V_m . To facilitate comparison between the two models, v was converted to the membrane potential: $V_m = V_{\text{rest}} + v(V_{\text{peak}} - V_{\text{rest}})$, where $V_{\text{rest}} = -81.2$ mV and $V_{\text{peak}} = 3.6$ mV.

The dynamics of the membrane model (CRN or FK3V) is captured in I_{ion} which is the sum of the individual ionic currents. The intracellular resistivity of 125 Ωcm was chosen to give a physiologic CV of approximately 60 cm/s. The fiber was 2 cm long and was stimulated at one end. The APD measurement location was in the middle, 1 cm away from both the stimulus site and the end of the fiber. This geometry ensured that at the measurement location the effects of the stimulus and the sealed opposite end were negligible and that the membrane potential waveform had reached steady-state.

Equation (1) was solved numerically using the Crank–Nicolson method² and gating variables were computed using the Euler method.³ In both models, a time step of 15 μs and a spatial step of 100 μm were used. The above steps were conservatively chosen such that activation threshold, action potential shape, and CV varied by less than 5% when either step was halved.

Determination of Restitution Curves

Activation time was taken as the time the membrane potential crossed -40 mV from below during the upstroke. Recovery time and therefore APD were taken as the time the membrane potential recovered to 93.1% (this value minimizes the effect of stimulus artifacts on the APD computation). To determine CV, activation times were measured at the APD measurement location and two sites placed 0.5 mm away on each side of the APD measurement location. The CV was computed as an inverse slope of the line that gave the best least-square fit to the three activation times.

Restitution was measured using the downsweep protocol by sequentially decrementing pacing period (referred to in this communication as basic cycle length or BCL). Steady-

state diastolic interval (DI), APD, and CV were measured at each BCL. The resulting DI/APD pairs and DI/CV pairs formed points on the restitution curves. In all cases, BCL was decremented from 1000 to 350 ms in steps of 50 ms and the number of stimuli (10–50) at each BCL was sufficient to reach steady state. The computation time for the entire APD and CV restitution curves (such as those shown in Fig. 1) was a few hours for the CRN fiber and tens of minutes for the FK3V fiber. Simulations can be accelerated by the use of the look-up tables and adaptive time stepping but these techniques were not used in this study.

First, a downsweep protocol was applied to the CRN fiber and APD and CV restitution curves were determined. Because the FK3V model does not account for variations in intracellular concentrations and to eliminate nonphysiologic drift,^{8,9} intracellular sodium and potassium concentrations were held constant in the CRN fiber. Next, the parameters of the FK3V model were chosen by repeatedly applying a downsweep to the FK3V fiber, comparing the resulting restitution curves with the CRN restitution curves, and adjusting the parameters for the next downsweep. The next section describes in detail the two-step procedure that was used to choose the parameters required to reproduce the APD and CV restitution of the CRN model.

Procedure for Choosing Parameters of the FK3V Model

Choosing parameters of the FK3V model consisted of two steps. Step 1 was designed to adjust specific parameters in an attempt to reproduce specific features of the restitution curves and obtain restitution curves close to those of the CRN fiber. Step 2 was designed to fine-tune all parameters to match the restitution curves as closely as possible.

Experience with the FK3V model, analysis of the model equations, and a procedure used by Fenton⁴ were used throughout Step 1. The following was found to be a useful sequence of actions for Step 1:

1. Adjust all parameters to approximate action potential shape from rest.

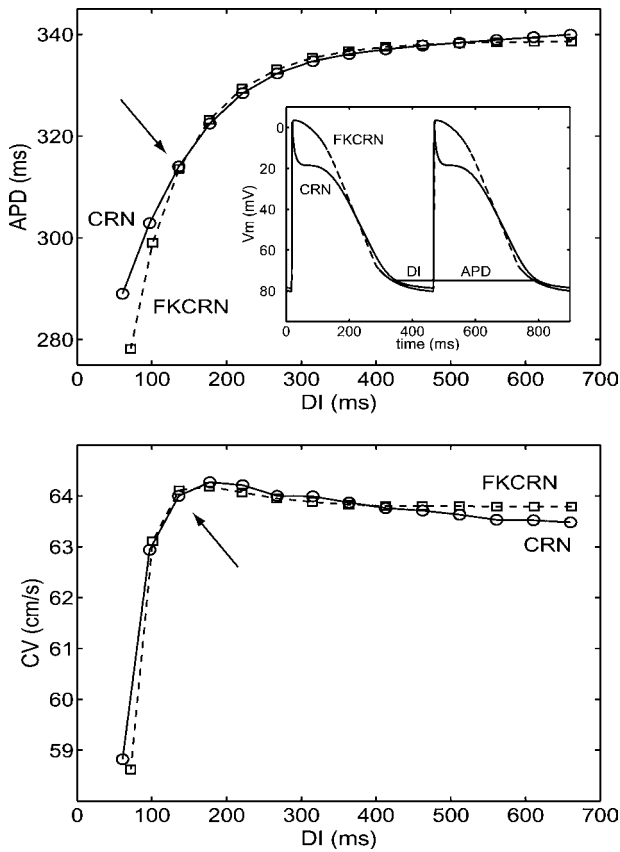


FIGURE 1. Restitution curves obtained from a fiber having CRN membrane dynamics (solid line) and a fiber having FKCRN membrane dynamics (dashed line). Top: APD restitution. Top inset: Action potentials obtained during pacing at a BCL of 450 ms. Bottom: CV restitution. Arrows show DI of 136 ms, APD of 314 ms, and CV of 64 cms corresponding to pacing at a BCL of 450 ms.

2. Adjust τ_{fast} to approximate maximum CV.
3. Adjust τ_{fopen1} , τ_{fopen2} , and τ_{fclose} to approximate minimum CV.
4. Adjust τ_{fopen1} and τ_{fopen2} to approximate minimum DI and CV curve shape at all DIs.
5. Adjust τ_{fopen1} to approximate CV curve shape at short DIs.
6. Adjust τ_{slow} to approximate maximum APD.
7. Adjust τ_{ung2} to approximate minimum APD.
8. Adjust τ_{ung1} to approximate APD curve shape at long DIs.

Step 1 demonstrated that a parameter could affect more than one restitution characteristic and a parameter could affect a restitution characteristic in an unexpected way. To assess the effect of varying a parameter, sensitivity graphs for each parameter were generated, which were then used to develop the sensitivity table shown as Table 2.

The sensitivity graphs and Table 2 were used as guides throughout Step 2. The following was found to be a useful

sequence of actions for Step 2:

1. Adjust the fast-current parameters to obtain a closer match of the CV restitution.
2. Adjust the slow and ungated current parameters to obtain a closer match of the APD restitution.
3. Adjust all parameters to match CV and APD restitution as closely as possible.

When adjusting for CV (APD), APD (CV) restitution was monitored for unintended effects and to ensure APD (CV) restitution was not changed significantly.

RESULTS

Table 1 lists the parameter values chosen to reproduce the CRN restitution curves; the FK3V model with these parameters is hereafter called the FKCRN model. Figure 1 shows the restitution curves for those parameters and the CRN model. The APD curves are virtually identical down to a DI of about 135 ms. The CV curves match closely up to a DI of about 500 ms. Despite numerous attempts the slight deviations in APD at short DI and slight deviations in CV at long DI could not be reduced without unacceptably increasing deviations at other DIs. The deviation in APD at short DI is of more consequence because it can potentially affect the dynamic behavior of the model. However, for neither model the maximum slope of the restitution curve is close to unity (0.38 for CRN and 0.71 for FKCRN) so we do not expect significant differences in the dynamic behavior of the two models. In addition, the APD restitution curve measured in human atrium⁷ is steeper than that of the CRN model. Thus, the APD restitution curve of the FKCRN model was deemed acceptable.

The inset in the top panel of Fig. 1 shows action potentials during pacing at a BCL of 450 ms in the CRN and FKCRN fibers. The DI and APD labeled on the inset plots of the top panel are 136 and 314 ms, respectively, and are indicated on the restitution curves with arrows. In addition to having the same DI and APD, the upstrokes were very similar. While the plateau of the FKCRN action potential was of appropriate duration, its morphology differed from the CRN action potential. This was not surprising since the FK3V model lacks the transient outward and ultrarapid rectifier potassium currents of the CRN model that are responsible for the rapid repolarization occurring just after the upstroke. A closer match of the action potential shape can be achieved using the four-variable model recently proposed by Fenton.⁶ However, the four-variable model has 22 parameters instead of 13, which would complicate the matching procedure.

Table 2 summarizes the effects of each parameter of the FKCRN model on the restitution curves. In constructing this table, changes less than a few milliseconds or tenths of a cm/s are classified as slight or no effect even though all parameter variations had some effect. Review of Table 2

TABLE 2. Sensitivity table for Fenton–Karma model parameters.

Parameter	DI _{min}	APD curve	APD _{min}	APD _{max}	CV curve	CV _{min}	CV _{max}
τ_{fast}	↑	—	—	—	Lowers	↓	↓
$\tau_{f\ open1}$	↑	—	↓	—	SNC↓	↓	—
$\tau_{f\ open2}$	↑	—	↓	—	SNC↓	↓	—
$\tau_{f\ close}$	↓	Shifts L/U	↓	↑	Raises	↓	↑
V_f	—	CCW	—	↑	Lowers	↓	↓
$V_{f\ open}$	—	—	—	—	—	—	—
τ_{slow}	—	CW	—	↓	—	—	—
τ_{sopen}	—	CW	↓	↓	—	—	—
τ_{sclose}	—	CCW	—	↑	—	—	—
V_{slow}	—	Lowers	↓	↓	—	—	—
K	—	Raises	—	↑	—	—	—
τ_{ung1}	↑	CCW	↓	↑	SNC↑	↑	↑
τ_{ung2}	—	Raises, CCW	↑	↑	—	↓	—

Note. Effect of increasing parameter on values listed in each column: —: slight or no effect; ↓: value decreased; ↑: value increased; ↓: effect mixed; shifts L/U: shifts curve left and up; CW: rotates curve clockwise; CCW: rotates curve counter-clockwise; SNC↓: supernormal conduction decreased; SNC↑: supernormal conduction increased. (Supernormal conduction occurs when CV increases with decreasing DI.)

reveals three general observations: (1) the fast-current parameters have significant effect on minimum DI and CV restitution and moderate effect on APD restitution; (2) the slow-current parameters have significant effect on APD restitution and very little effect on minimum DI and CV restitution; and (3) the ungated-current parameters have significant effect on APD restitution and moderate effect on minimum DI and CV restitution. Some effects were asymptotic in that after reaching a certain parameter value any further variation in that parameter no longer had any consequence. Also, effects on the restitution curves sometimes varied with DI, and the effects of some parameters were enhanced or diminished as a function of other parameters. Nevertheless, observations drawn from Table 2 are consistent with the comments in the original publication of the FK3V model, indicating that the qualitative effects of the parameters are fairly independent of the source of the restitution curves used for matching. Thus, Table 2 provides detailed insight into the behavior of the FK3V model and the effect of FK3V model parameters on its restitution properties.

DISCUSSION

This communication presents a procedure used to reproduce the APD and CV restitution properties of the CRN model using the FK3V model. The resulting three-variable FKCRN model closely matches the restitution properties of the more complex CRN model, at a small fraction of the computational expense. Even though the FKCRN model does not reproduce exactly the shape of the action potential or effects related to changes in intracellular concentrations, its simplicity and computational efficiency make it a useful tool in large-scale simulations of atrial arrhythmias.

As with every empirical procedure, ours is to some extent subjective and the resulting parameter set may not be optimal or unique. One may argue for an automated procedure in which a least-square fit would determine all 13 parameters at once. However, an automated procedure would have a hard time dealing with multiple minima or with large regions in parameter space that give a fit of essentially the same quality. Both problems are likely to occur in a high-dimensional parameter space, as evidenced by Table 2, which shows that several parameters have little or no effect on restitution properties. Thus, we opted for an empirical procedure that is based on the understanding of the model.

Such an understanding is provided by Table 2 that can guide the matching process. This sensitivity table was constructed for the parameter set determined in Step 1, so it might be different in a different portion of the parameter space than that investigated. We believe that any such differences will be minor: cardiac restitution curves are fairly similar and thus the fitting procedure is likely to operate in the same portion of the parameter space regardless of the model or experiment that provided the curves. This conjecture is supported by the fact that Table 2 agrees with the observations on the role of parameters made by Fenton and Karma, who matched four different restitution curves.⁵ In addition, parameters chosen here for the FKCRN model and the four sets of parameters in the original publication⁵ are generally of the same order of magnitude. The only exception is τ_{fopen2} (τ_{v1} in Ref. 5), which takes values from 18.2 to 1250. Thus, there is a reason to expect that the procedure and the sensitivity table presented in this communication are general and could be used to match other membrane models and experimental results.

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